

Real-Time Emotion Detection from Text for Expressive Avatar Control in Conversational AI Systems

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Abstract—This paper presents the design, implementation, and evaluation of a real-time emotion detection module for conversational AI assistants. Given the text generated by a large language model (LLM) in streaming mode, the module classifies the predominant emotion and distributes the result to two downstream components: an avatar controller for visual expression and a text-to-speech engine for vocal tone adaptation. We fine-tune `distilbert-base-multilingual-cased`, a six-layer multilingual DistilBERT, on a bilingual dataset combining GoEmotions (English, 58k examples) and its Spanish machine translation (54k examples), collapsing the original 28-class taxonomy into 10 semantically coherent classes. The model is exported to ONNX with INT8 quantization for local inference on commodity hardware. On the combined test set ($n=10,824$) it attains a macro F1 of 0.544; an INT8-vs-FP32 comparison shows the quantization is effectively lossless (≈ 0 pp on held-out data) while running 3–4 $\times$ faster. End-to-end streaming latency is a P95 of 3.1ms (single forward pass ≈ 6 ms at 128 tokens), an order of magnitude under the sub-80ms target for imperceptible delay before speech synthesis. On a 60-scenario bilingual manual evaluation (10 emotions $\times$ 2 languages $\times$ 3 lengths) the module reaches 60.0% accuracy on the held-out test split and 66.7% pooled.

Index Terms—emotion recognition, natural language processing, conversational AI, real-time inference, ONNX, multilingual, avatar control

I. INTRODUCTION

Conversational AI assistants have become increasingly capable of generating coherent and contextually appropriate responses. However, the gap between linguistic competence and expressive delivery remains significant. Users perceive assistants as more natural and engaging when vocal tone and visual cues (such as facial expressions of a virtual avatar) are congruent with the emotional content of the message being delivered [1].

This work presents the Emotion Detection Module (Module A), a component of a larger holographic AI assistant system designed to run entirely on-device without cloud dependencies. The module addresses the following problem: *given a stream of text generated by an LLM, classify the predominant emotion in real time, with latency low enough that the result is available before the text-to-speech engine begins synthesis.*

The main contributions of this paper are:

- A bilingual (English/Spanish) fine-tuned `distilbert-base-multilingual-cased` classifier for 10-class emotion detection from short conversational text.
- An empirically motivated class collapse scheme reducing GoEmotions’ 28 classes to 10, with decisions grounded in co-occurrence analysis and qualitative inspection.
- A streaming inference architecture (`push/flush` interface) with exponential moving average smoothing, designed for integration into real-time LLM pipelines.
- An ONNX INT8 quantization-cost analysis showing the quantized model matches FP32 accuracy (≈ 0 pp on held-out data) while running 3–4 $\times$ faster on CPU.
- A latency benchmark validating the sub-80ms P95 target with an order-of-magnitude margin, and a split-disciplined 60-scenario bilingual streaming evaluation.

II. RELATED WORK

A. Emotion Recognition from Text

Emotion recognition in natural language has been approached through lexicon-based methods [2], traditional machine learning [3], and more recently, transformer-based models [4]. The introduction of GoEmotions by Demszky et al. [5] provided the first large-scale fine-grained emotion dataset for English text, enabling fine-tuning of pre-trained language models for this task.

B. Multilingual Emotion Recognition

Cross-lingual emotion detection remains an open challenge. XLM-RoBERTa [6] has demonstrated strong cross-lingual transfer on a variety of NLP tasks. However, training data for emotion recognition in languages other than English remains scarce. Translation-based dataset augmentation has been proposed as a practical alternative [7].

C. Affective Computing for Virtual Agents

The alignment of verbal and non-verbal cues in virtual agents has been studied in the context of embodied conversational agents [8]. Audiovisual coherence—where vocal tone matches facial expression—has been shown to increase user trust and perceived naturalness [9].

Figura 1 — Distribución de clases en GoEmotions
Las clases en rojo presentan desbalance severo respecto a la media

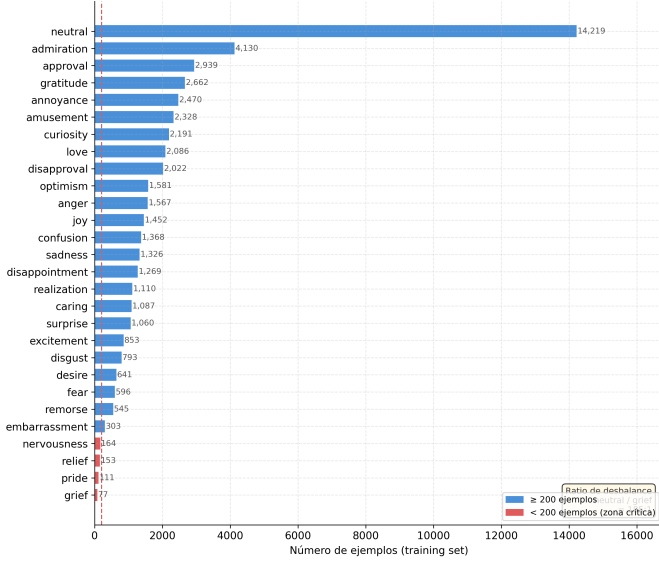


Fig. 1. Raw GoEmotions label distribution over the 28 original classes, showing the strong imbalance (*neutral* dominant, several minority emotions) that motivates collapse.

TABLE I
CLASS COLLAPSE SCHEME (28 → 10)

Final Class	GoEmotions Classes Included
joy	joy, amusement, excitement
love	love, caring, gratitude
admiration	admiration, approval, pride
optimism	optimism, relief, desire
curiosity	curiosity, realization, surprise, confusion
sadness	sadness, grief, remorse, disappointment
anger	anger, annoyance, disapproval
disgust	disgust
fear	fear, nervousness, embarrassment
neutral	neutral

III. METHODOLOGY

A. Dataset and Class Collapse

We use GoEmotions [5], a dataset of 58,009 Reddit comments annotated with 28 emotion categories whose raw label distribution is strongly imbalanced (Fig. 1). To address the needs of a conversational assistant context—where fine-grained distinctions between semantically similar emotions may not be reliably detectable or actionable—we collapse the 28 classes into 10 using empirical evidence from two sources:

Co-occurrence analysis. We compute the conditional probability matrix $P(j|i)$ over the multi-label raw annotations and identify pairs with high mutual co-occurrence as candidates for collapse (Fig. 2).

Qualitative inspection. For each candidate pair, we inspect representative examples from the dataset to confirm that the semantic distinction is meaningful in a conversational LLM context.

The resulting 10-class scheme is shown in Table I.

Figura 4 — Co-ocurrencia de etiquetas ($P(j|i)$, versión multi-label)
Bloques de alta co-ocurrencia sugieren candidatos a colapso de clases

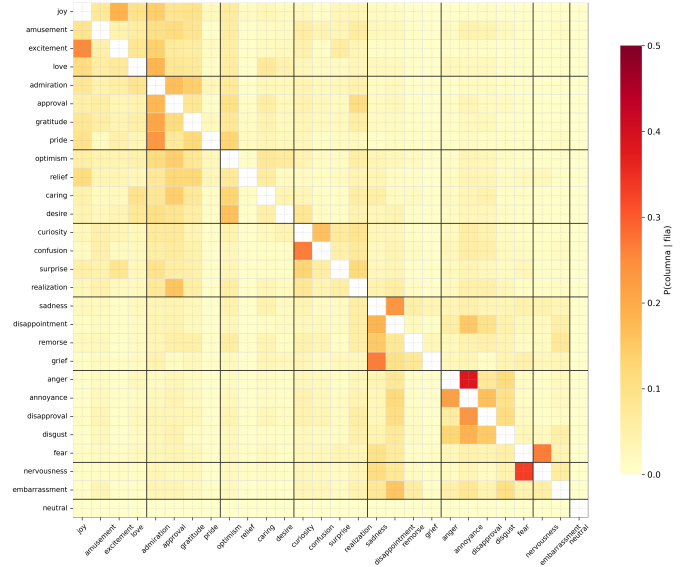


Fig. 2. Label co-occurrence heatmap over the raw multi-label annotations. High mutual co-occurrence between semantically adjacent emotions guides the 28→10 collapse.

Figura 5 — Distribución tras colapso de clases (10 grupos)
Ratio de desbalance reducido de 185:1 → 16.1:1

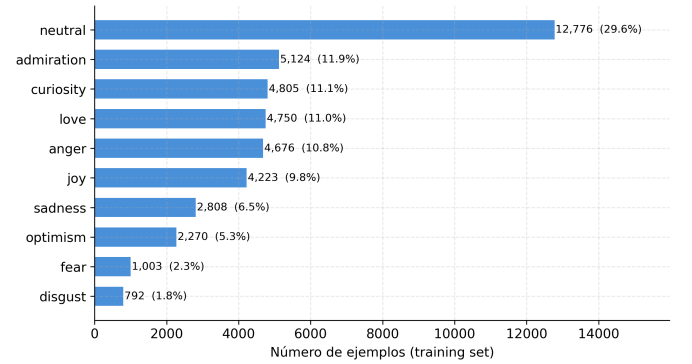


Fig. 3. Class distribution after the 28→10 collapse. *neutral* remains the majority class, reflecting its prevalence in the Reddit-sourced data.

Note that *pride* was initially kept as an independent class (12-class scheme) due to its relevance for eccentric assistant personalities. However, a first training run yielded $F1 = 0.364$ for *pride*, with 38% of true *pride* instances predicted as *admiration*. Qualitative analysis confirmed that *pride* in GoEmotions predominantly captures *admiring pride* (“I’m so proud of my parents”) rather than egocentric arrogance—semantically indistinguishable from *admiration* in this dataset. No NLP dataset with egocentric *pride* as a labeled class exists in standard taxonomies (Ekman, Plutchik, Shaver). *Pride* was therefore collapsed into *admiration*. The class distribution after collapse is shown in Fig. 3.

To extend coverage to Spanish, we augment the dataset with `mrm8488/go_emotions-es-mt` [10], a machine-translated version of GoEmotions with 54,263 examples, iden-

TABLE II
FINAL TRAINING HYPERPARAMETERS

Parameter	Value
Base model	distilbert-base-multilingual-cased
Max sequence length	128
Batch size	32
Epochs	4 (early stopped; best epoch 4)
Learning rate	2e-5
Warmup ratio	0.10
Weight decay	0.05
Gradient clipping	1.0
ES patience	3 (monitoring F1-macro)
Seed	42

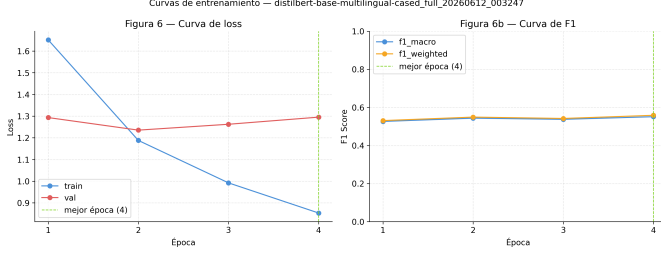


Fig. 4. Training and validation loss and F1-macro across epochs for the shipped run. The best checkpoint (epoch 4) is selected by validation F1-macro.

tical class schema, and the same train/val/test splits. The combined dataset totals approximately 112k examples in a 50/50 EN/ES ratio.

B. Model Architecture

We fine-tune **distilbert-base-multilingual-cased** for sequence classification: a six-layer distilled transformer pre-trained on 104 languages, providing genuine multilingual representations in a compact footprint. We also trained an XLM-RoBERTa-base variant [6] as a stronger-encoder reference. The multilingual DistilBERT was shipped because it delivers comparable macro-F1 to the larger model at a fraction of the parameter count and inference cost—decisive for the sub-80ms, on-device target where the model must run on a commodity CPU alongside the rest of the assistant stack. A larger encoder offered no accuracy gain that would justify its latency and memory overhead in this deployment.

Class imbalance is addressed via `compute_class_weight='balanced'` from `scikit-learn`, applied as sample weights to `nn.CrossEntropyLoss`.

C. Training Configuration

Training was performed on an NVIDIA RTX 4060 Ti (16GB VRAM) using PyTorch. Table II summarizes the final hyperparameters; all random seeds were fixed at 42 for reproducibility. Training converged in four epochs under F1-macro-monitored early stopping (patience 3). The resulting training curves are shown in Fig. 4.

D. ONNX Export and Quantization

The trained model is exported to ONNX FP32 with Hugging Face Optimum [11], via `ORTModelForSequenceClassification` with `export=True`. Dynamic INT8 quantization is then applied offline with `onnxruntime.quantization`’s `quantize_dynamic`, quantizing weights to INT8 while performing activation quantization at runtime. Prediction consistency between PyTorch, ONNX FP32, and ONNX INT8 was verified on the full test set before deployment.

E. Streaming Inference Architecture

The `EmotionDetector` class implements a push/flush interface for real-time streaming integration:

- `push(chunk: str)`: accumulates incoming text. Inference is triggered when the buffer reaches `min_tokens=128` (the shipped default) near a sentence boundary, or when the buffer reaches `min_tokens * 2` regardless of punctuation.
- `flush()`: forces final inference at end of LLM turn; clears buffer and resets EMA state. `flush()` always emits, bypassing the mid-turn confidence gate.

A `confidence_floor` of 0.65 suppresses low-confidence mid-turn emissions, so the avatar and TTS engine only react once the prediction has stabilized; `flush()` is exempt from this floor.

To smooth rapid emotion fluctuations across chunks, we apply an exponential moving average over class probability vectors:

$$\hat{p}_t = \alpha \cdot p_t + (1 - \alpha) \cdot \hat{p}_{t-1} \quad (1)$$

where $\alpha = 0.40$ was calibrated empirically. The final classification is taken as $\arg \max(\hat{p}_t)$ subject to per-class logit bias (default: none) and confidence thresholds, both configurable per personality via `set_personality()` without retraining. The shipped model path uses `thresholds={}`; the single fixed per-class threshold for *disgust* (0.68) lives downstream in the Output Formatter rather than in the model (see §V).

IV. RESULTS

A. Classification Performance

Table III reports precision, recall, and F1 per class on the combined EN/ES test set (n=10,824). The macro-averaged F1 of 0.544 compares favorably against the GoEmotions baseline of 0.46 reported by Demszky et al. [5] with BERT-base over the full 28-class taxonomy, noting that our 10-class collapse reduces task difficulty. Per-class F1 and the test-set confusion matrix are shown in Figs. 5 and 6.

The low recall for *neutral* (0.38) is consistent with the intended design: an assistant with an expressive personality should emit *neutral* less frequently than its distribution in Reddit-sourced data.

TABLE III
PER-CLASS RESULTS ON COMBINED TEST SET (N=10,824)

Class	P	R	F1	Support
joy	0.63	0.72	0.67	951
love	0.75	0.73	0.74	1228
admiration	0.53	0.56	0.55	1295
optimism	0.45	0.61	0.52	539
curiosity	0.51	0.65	0.57	1212
sadness	0.45	0.57	0.50	650
anger	0.44	0.53	0.48	1238
disgust	0.36	0.54	0.43	246
fear	0.42	0.61	0.50	264
neutral	0.68	0.38	0.49	3201
macro avg	0.52	0.59	0.54	10824
weighted avg	0.58	0.55	0.55	10824

Figura 7 — F1 por clase en test set
F1 macro = 0.5445 (entrenamiento completo)

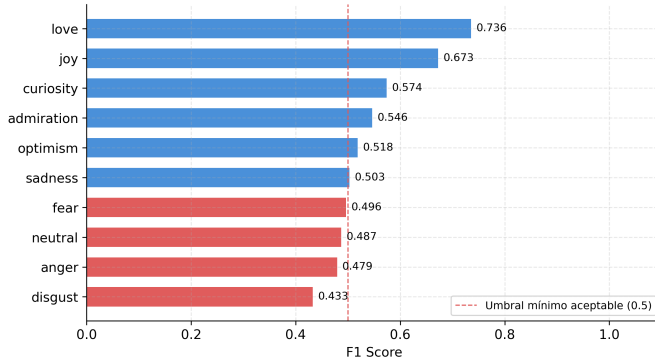


Fig. 5. Per-class F1 on the combined EN/ES test set. *love* and *joy* are strongest; *disgust* (a single source label) is weakest.

Figura 8 — Confusion matrix normalizada (test set)
Diagonal = correctas | Off-diagonal = confusiones del modelo

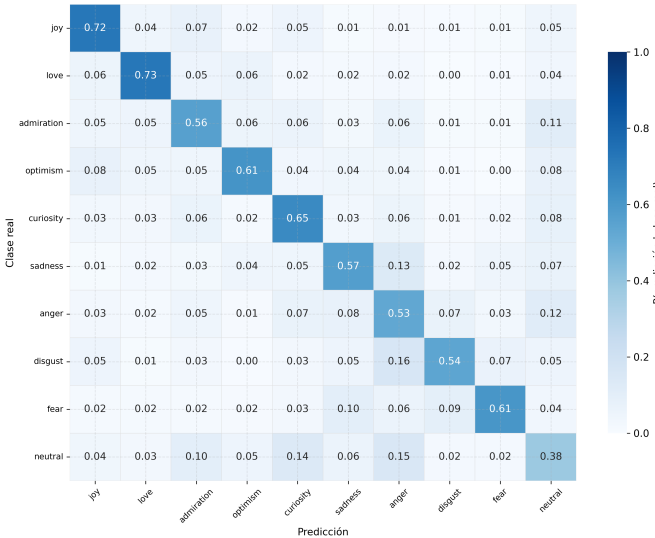


Fig. 6. Confusion matrix on the combined test set. Confusions concentrate among semantically adjacent emotions and into *neutral*.

B. Latency Benchmark

Table IV reports P50/P95/P99 latency of a single forward pass at four fixed sequence lengths (20 warmup + 100 timed

TABLE IV
SINGLE-PASS LATENCY BY SEQUENCE LENGTH (ms, CPU)

Tokens	INT8			FP32		
	P50	P95	P99	P50	P95	P99
16	1.7	2.5	5.5	6.9	8.2	9.4
32	2.4	3.0	3.4	7.8	8.8	9.2
64	2.9	3.2	3.3	9.2	10.2	11.0
128	5.2	6.0	6.3	15.6	16.9	17.2

Figura 9 — Benchmark de latencia por configuración y longitud de chunk

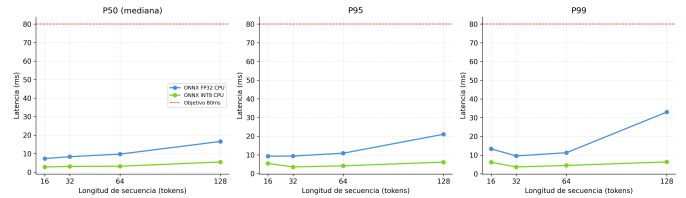


Fig. 7. Single-pass latency vs. sequence length for the ONNX INT8 and FP32 models on CPU.

Figura 10 — Speedup relativo a ONNX FP32 CPU (P50)

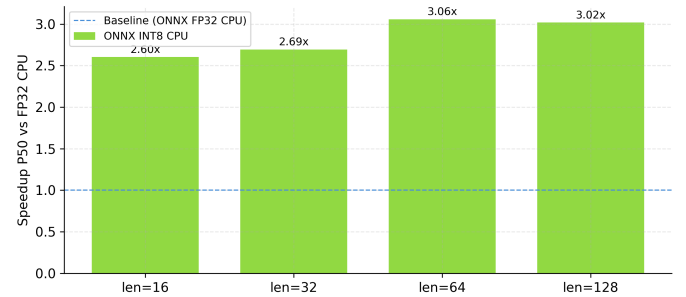


Fig. 8. INT8 speedup over FP32 by sequence length: a consistent 3.0–4.1× reduction in CPU latency.

runs each, dynamic padding, `CPUExecutionProvider`). Both the INT8 production model and the FP32 reference are measured; Figs. 7 and 8 visualize the comparison and the resulting speedup. GPU execution via `CUDAExecutionProvider` is used automatically when available; the reported numbers are the CPU-only worst case.

INT8 quantization yields a consistent 3.0–4.1× speedup over FP32 on CPU. Measured end-to-end through the streaming detector (push→flush→emit, 200 runs), the INT8 path has a P95 of 3.1ms versus 9.6ms for FP32. Even the slowest configuration (FP32 at 128 tokens, P95 16.9ms) is well within budget; the shipped INT8 path clears the sub-80ms target by more than an order of magnitude.

This delay represents a *start delay*—the TTS engine waits for the first `EmotionEvent` before beginning synthesis—rather than simultaneous audiovisual desynchronization. The literature reports audiovisual synchrony thresholds of 20–49ms for simultaneous stimuli [12], which do not directly apply to this pre-synthesis delay scenario.

TABLE V
STREAMING EVALUATION ACCURACY (INT8, LOCKED CONFIG)

Split	n	Acc.	EN	ES
test_final (held-out)	20	60.0%	60%	60%
dev	40	70.0%	75%	65%
pooled	60	66.7%	70%	63%

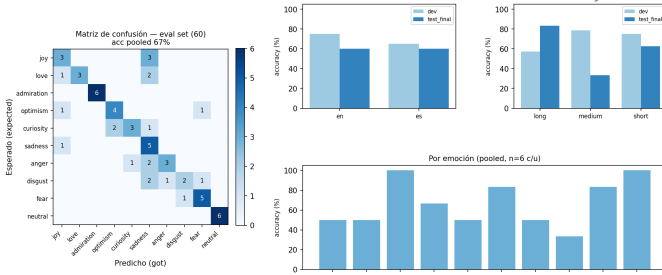


Fig. 9. Manual streaming evaluation of the active INT8 model: confusion matrix and accuracy breakdowns by language and length.

C. Bilingual Streaming Evaluation

To validate the model end-to-end through the production EmotionDetector (text pushed word-by-word, then `flush()`), we built a manual evaluation set of **60 scenarios** spanning 10 emotions $\times$ 2 languages (EN/ES) $\times$ 3 lengths (short/medium/long). The set is split into a development partition (40 scenarios) and a held-out `test_final` partition (20). To avoid optimistic bias, we enforce split discipline: a hyperparameter sweep over `min_tokens` $\in$ {64, 128}, `ema_alpha` $\in$ {0.30, 0.40, 0.50} and a neutral logit bias was run *on dev only*, and `test_final` was measured *once* with the locked shipped configuration (`min_tokens=128`, `ema_alpha=0.40`, no logit bias).

The held-out result (60.0%) confirms the pooled estimate (66.7%). The dev sweep moved accuracy by at most one scenario out of 40 (within noise) and the neutral logit bias gave no gain—neutral recall was already 100% with zero false positives on dev—so the shipped configuration was kept unchanged, i.e. not tuned to the 40-sample dev split. Per emotion (pooled, $n=6$ each), *admiration* and *neutral* reach 100%, *sadness* and *fear* 83%, *optimism* 67%, while *joy*, *love*, *curiosity* and *anger* sit at 50% and *disgust* is weakest at 33%. The dominant error sink is *sadness* (joy/love/anger leak into it, mostly on Spanish); *disgust* is limited by being the only emotion backed by a single GoEmotions source label. The per-split confusion matrices and breakdowns are shown in Fig. 9.

Statistical caveat. With $n = 20$, the `test_final` estimate is wide: a Wilson 95% confidence interval for 60.0% is $\approx$ [39%, 78%], and for the pooled 66.7% ($n = 60$) it is $\approx$ [54%, 77%]. We therefore treat `test_final` as a held-out *confirmation* and lean on the pooled set and the GoEmotions macro-F1 (0.544, $n = 10,824$) for the precise quality claims, rather than over-interpreting 20 samples.

Label note. The scenario `test_es_love_medium` (a

TABLE VI
QUANTIZATION COST: FP32 VS. INT8 (SAME LOCKED CONFIG)

Split	INT8	FP32	Agreement
test_final	60.0%	60.0%	90%
dev	70.0%	72.5%	95%
pooled	66.7%	68.3%	93%

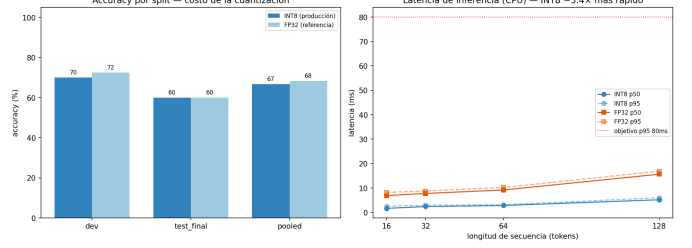


Fig. 10. Quantization cost and latency: FP32 vs. INT8 accuracy per split alongside the INT8 latency advantage.

deceased grandmother) is a mixed love/grief case; *love* was retained after manual review because the dominant frame is gratitude/affection. This decision is documented in the results JSON.

D. Quantization Cost (INT8 vs. FP32)

Because the production model is INT8, we quantify what quantization costs by running the same locked configuration with the FP32 reference on the identical evaluation set (Table VI, Fig. 10).

INT8 costs **0 pp** on the held-out set and at most one scenario elsewhere (dev +2.5 pp, pooled +1.6 pp—both within noise), with 90–95% prediction agreement. Combined with the 3–4 $\times$ CPU speedup of §IV-B, this justifies shipping INT8 as the production artifact at no measurable accuracy cost.

V. DISCUSSION

Domain shift. GoEmotions is sourced from Reddit, which skews toward short, emotionally explicit text. LLM-generated conversational responses tend to be more narrative and tonally subtle. This explains the degradation to neutral on longer, less explicit texts. Future work should include fine-tuning on LLM-generated synthetic data with controlled emotional content.

Disgust threshold placement. *disgust* is the weakest class—backed by a single GoEmotions source label and rarely surpassing anger in production text even when expressing clear rejection. Rather than bake a per-class threshold into the model, the model path ships with `thresholds={}`: the classifier emits a calibrated probability vector unchanged. The single fixed *disgust* threshold of 0.68 instead lives downstream in the Output Formatter, keeping the model artifact policy-free and the emission rule auditable in one place. This separation lets the threshold be tuned per deployment without touching the model.

Personality system. The `set_personality()` interface enables runtime configuration of per-class logit bias and

confidence thresholds without model retraining. This allows deployment of eccentric assistant profiles (e.g., an irritable assistant with positive bias on anger) while maintaining a single model artifact.

VI. CONCLUSIONS

We presented a real-time emotion detection module designed for integration into a local, on-device conversational AI assistant. The module classifies 10 emotion classes from streaming text with an end-to-end P95 latency of 3.1ms (≈ 6 ms single forward pass at 128 tokens) on commodity CPU hardware, clearing the sub-80ms pre-synthesis budget by more than an order of magnitude. Bilingual support for English and Spanish is achieved through dataset augmentation with machine-translated GoEmotions data and fine-tuning of `distilbert-base-multilingual-cased`, which reaches a macro-F1 of 0.544 on the combined test set.

Key design decisions—class collapse, EMA smoothing, INT8 quantization, and the placement of the disgust threshold in the Output Formatter—are grounded in empirical evidence and documented for reproducibility. In particular, the INT8 quantization-cost analysis shows the quantized model matches FP32 accuracy (≈ 0 pp on held-out data) while running $3\text{--}4\times$ faster. The module is deployed as a standalone ONNX artifact with a clean `push/flush/on_emotion` interface, enabling integration with any downstream component without coupling to the training framework.

Future work includes fine-tuning on LLM-generated synthetic emotional data, strengthening the weakest classes (*disgust*, and the Spanish *sadness* error sink) with higher-quality non-translated Spanish data, and evaluation on a wider range of personality configurations.

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